# Mimic X Adapter — ATARI / X68000 Joystick

USB-MIDI in from a phone, 9-pin ATARI-standard joystick out. Also emulates a Mega Drive 6-button pad.

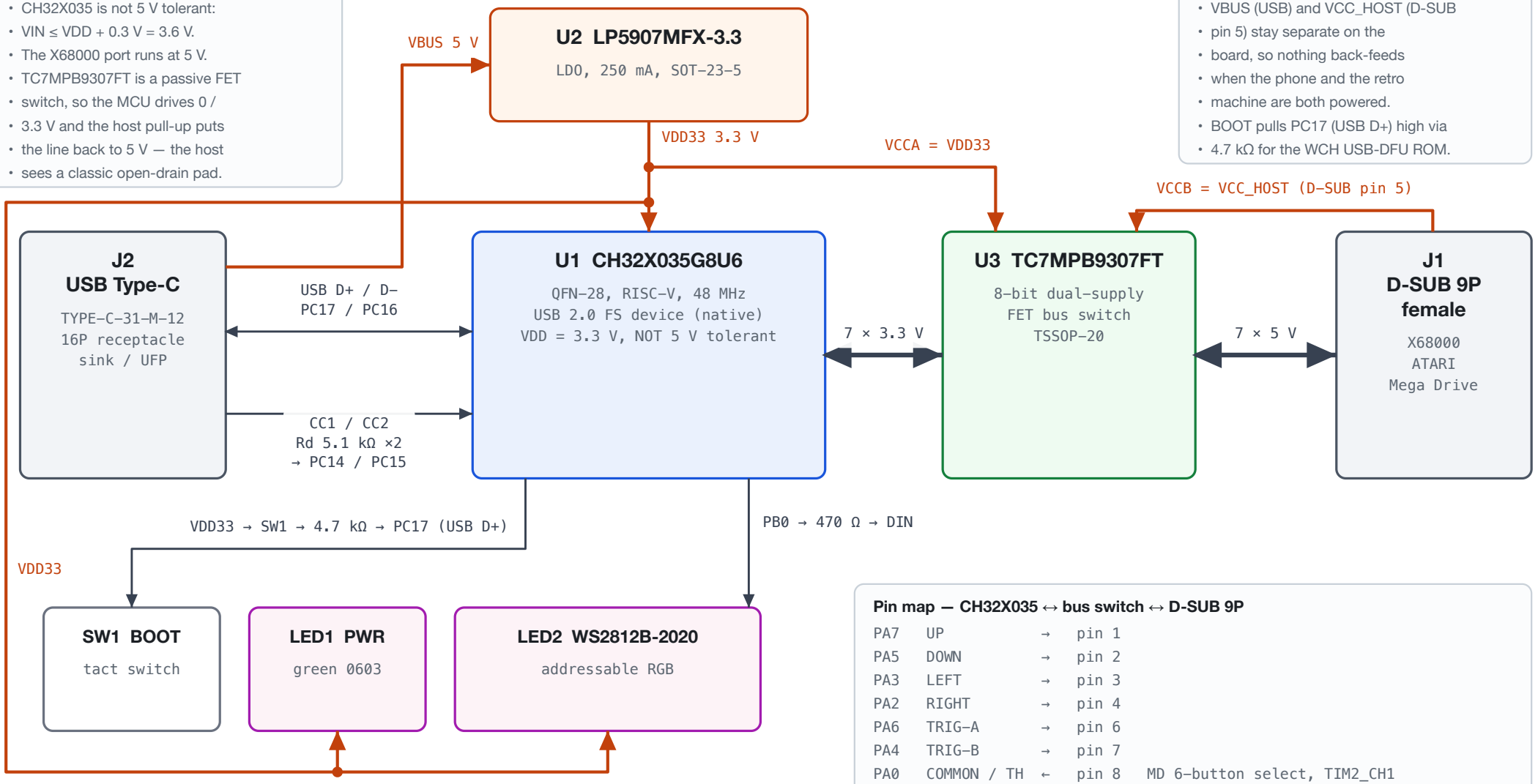
board rev 1.0.0 · CERN-OHL-S-2.0
USB VID:PID 0x1209:0x9210
github.com/kunichiko/MimicX-hardware

**Why the bus switch**

- CH32X035 is not 5 V tolerant:
- VIN ≤ VDD + 0.3 V = 3.6 V.
- The X68000 port runs at 5 V.
- TC7MPB9307FT is a passive FET
- switch, so the MCU drives 0 /
- 3.3 V and the host pull-up puts
- the line back to 5 V — the host
- sees a classic open-drain pad.

**Power domains**

- VBUS (USB) and VCC_HOST (D-SUB
- pin 5) stay separate on the
- board, so nothing back-feeds
- when the phone and the retro
- machine are both powered.
- BOOT pulls PC17 (USB D+) high via
- 4.7 kΩ for the WCH USB-DFU ROM.




**Pin map — CH32X035 ↔ bus switch ↔ D-SUB 9P**

| | | | |
|---|---|---|---|
| PA7 | UP | → | pin 1 |
| PA5 | DOWN | → | pin 2 |
| PA3 | LEFT | → | pin 3 |
| PA2 | RIGHT | → | pin 4 |
| PA6 | TRIG-A | → | pin 6 |
| PA4 | TRIG-B | → | pin 7 |
| PA0 | COMMON / TH | ← | pin 8 MD 6-button select, TIM2_CH1 |
| | +5 V / GND | | pin 5 / pin 9 |

power rail · signal · multi-bit bus

Every net shown here is declared in the .ato source; this drawing is generated from it by hand, not extracted.